Rajat M

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LinkedIn Profile

GitHub: https://github.com/rajat16127

📍 Mumbai, Maharashtra

Summary

- **Quantitative Research / ML** candidate with strong Python skills and hands-on experience in ML, deep learning, and quantitative modeling.
- Built and evaluated **quantitative trading, ML, and computer vision** systems using real-world datasets.
- Experienced with **time-series analysis**, backtesting, feature engineering, and metrics such as CAGR, Sharpe Ratio, and drawdown.
- Worked on YOLO-based object detection pipelines involving data collection, augmentation, model optimization, and Docker deployment.
- Interested in research-driven environments combining **ML, statistics, and financial markets**.

Skills

- **Programming:** Python, JavaScript
- **Quant & ML:** Time-series analysis, feature engineering, regression, classification, backtesting, model evaluation
- **Libraries:** Pandas, NumPy, scikit-learn, TensorFlow, Keras, OpenCV, Ultralytics YOLO
- **Deep Learning:** CNNs, transfer learning, object detection, data augmentation
- **Tools:** Git, Docker, Linux, Jupyter Notebook, Matplotlib

Experience

AI/ML Engineer — BitVivid Solutions (2026 FEB – Present)

- Trained and optimized YOLO object detection models on custom datasets for real-time video analytics tasks.
- Worked on data collection, preprocessing, augmentation, inference pipelines, and model accuracy improvements.
- Built Python-based video detection/counting systems and performed model deployment using Docker and OpenVINO.

Projects

Reversion-Based Quantitative Strategy📈(Project Report) https://rajat16127.github.io/home/quant_2.html

Developed a reversal-based equity strategy on BSE Sensex stocks using rolling out-of-sample validation. Achieved 18.73% CAGR with Sharpe Ratio 0.92 over 10 years after transaction costs.

- **Concepts:** Look-ahead bias prevention, validation splits, signal evaluation, portfolio rebalancing
- **Metrics:** CAGR, volatility, Sharpe ratio, drawdown, turnover
- **Tech:** Python, Pandas, NumPy, yfinance

Deep Learning Fashion Image Classification 👕🧢 (Project Report) https://rajat16127.github.io/home/image_dl.html

Built an image classification pipeline using CNNs and transfer learning (Xception) on ~3,800 images across 10 classes with augmentation and confusion-matrix evaluation.

- **Tech:** Python, TensorFlow, Keras
- **Focus:** Model training, generalization, evaluation

Home Loan Default Prediction Using ML 📊 (Project Report) https://rajat16127.github.io/home/loan_ml.html

Built ML models to predict loan default risk on 300k+ records with 120+ features using preprocessing, feature engineering, and model comparison.

- **Tech:** Python, Pandas, NumPy, scikit-learn
- **Focus:** EDA, preprocessing, evaluation

Education

Bachelor of Engineering (Information Technology)

University of Mumbai

Relevant coursework: Data Structures, Algorithms, Machine Learning, Artificial Intelligence, Data Mining, Cloud Computing, Distributed Systems.